# <네트워크 프로그래밍 실습 수행평가>

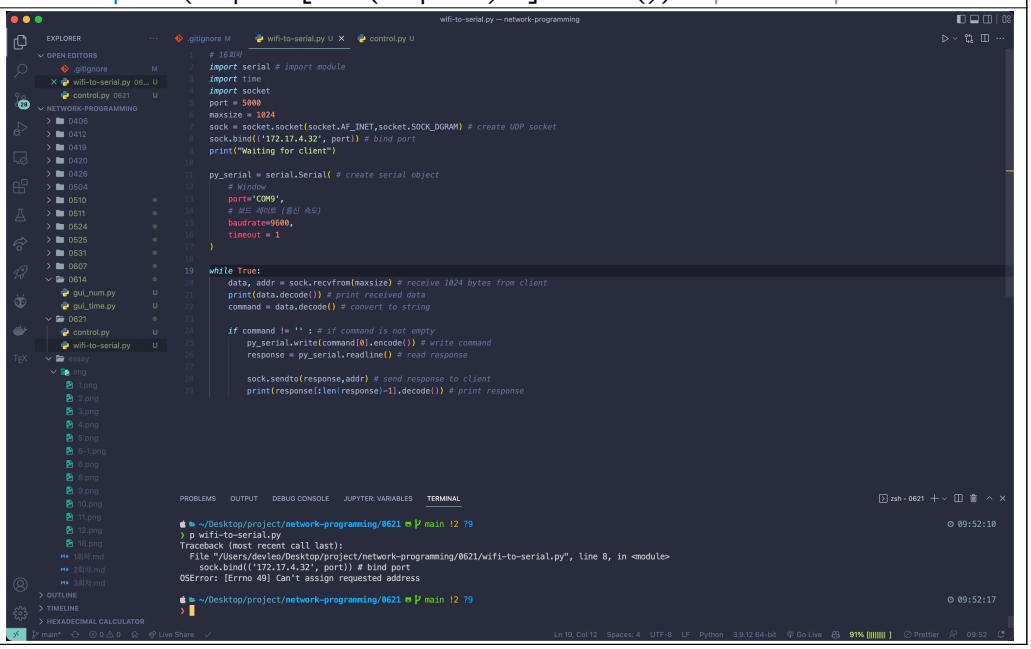
| 학번 | 3321 | 이름 | 배상혁 |
|---|---|---|---|
| 제출 날짜 | 2022.06.21 | | |
| 조원 이름 | | | |
| 실습 주제 | wifi to serial 분석 | | |

코드 전문

```python
# 16회차
import serial # import module
import time
import socket
port =5000
maxsize =1024
sock = socket.socket(socket.AF_INET,socket.SOCK_DGRAM) # create UDP socket
sock.bind(('172.17.4.32', port)) # bind port
print("Waiting for client")
py_serial = serial.Serial( # create serial object
    # Window
    port='COM9',
    # 보드 레이트 (통신 속도)
    baudrate=9600,
    timeout =1
)
while True:
    data, addr = sock.recvfrom(maxsize) # receive 1024 bytes from client
    print(data.decode()) # print received data
    command = data.decode() # convert to string
    if command !='' : # if command is not empty
        py_serial.write(command[0].encode()) # write command
        response = py_serial.readline() # read response

        sock.sendto(response,addr) # send response to client
        print(response[:len(response)-1].decode()) # print response
```

실행 이미지 파일



# <네트워크 프로그래밍 실습 수행평가>

| 학번 | 3321 | 이름 | 배상혁 |
|---|---|---|---|
| 제출 날짜 | 2022.06.21 | | |
| 조원 이름 | | | |
| 실습 주제 | built in LED on/off, LED on/off, 서보 모터 제어, GUI 구현 | | |

코드 전문

```
# 17~20회차
from tkinter import *
import socket
host ="172.17.4.32"
port =5000
sock = socket.socket(socket.AF_INET,socket.SOCK_DGRAM)
def button1_command():
    global sock, btn1_text, btn1_color

    if btn1_text =="ON":
        btn1_text ="OFF"
        sock = socket.socket(socket.AF_INET,socket.SOCK_DGRAM)
        sock.sendto("b".encode(),(host, port))
    elif btn1_text =="OFF":
        btn1_text ="ON"
        sock = socket.socket(socket.AF_INET,socket.SOCK_DGRAM)
        sock.sendto("a".encode(),(host, port))

    built_in_LED_button.configure(text=btn1_text, bg=btn1_color)
def button2_command():
    global sock, btn2_text, btn2_color

    if btn2_text =="ON":
        btn2_text ="OFF"
        sock = socket.socket(socket.AF_INET,socket.SOCK_DGRAM)
        sock.sendto("f".encode(),(host, port))
    elif btn2_text =="OFF":
        btn2_text ="ON"
        sock = socket.socket(socket.AF_INET,socket.SOCK_DGRAM)
        sock.sendto("e".encode(),(host, port))

    LED_button.configure(text=btn2_text, bg=btn2_color)
def button3_command():
    global sock, btn3_text, btn3_color

    if btn3_text =="ON":
        btn3_text ="OFF"
        sock = socket.socket(socket.AF_INET,socket.SOCK_DGRAM)
        sock.sendto("d".encode(),(host, port))
    elif btn3_text =="OFF":
        btn3_text ="ON"
        sock = socket.socket(socket.AF_INET,socket.SOCK_DGRAM)
        sock.sendto("c".encode(),(host, port))

    servo_button.configure(text=btn3_text, bg=btn3_color)
root = Tk()
btn1_color ="black"
btn2_color ="black"
```

```
btn3_color ="black"
btn1_text ="OFF"
btn2_text ="OFF"
btn3_text ="OFF"
built_in_LED_label = Label(text="Built in LED")
built_in_LED_button = Button(text=btn1_text, fg="black",
                    bg=btn1_color, command=button1_command)
built_in_LED_label.grid(row=0, column=0)
built_in_LED_button.grid(row=0, column=1)
LED_label = Label(text="LED")
LED_button = Button(text=btn2_text, fg="black",
                    bg=btn2_color, command=button2_command)
LED_label.grid(row=1, column=0)
LED_button.grid(row=1, column=1)
servo_label = Label(text="Servo")
servo_button = Button(text=btn3_text, fg="black",
                    bg=btn3_color, command=button3_command)
servo_label.grid(row=2, column=0)
servo_button.grid(row=2, column=1)
root.mainloop()
```

실행 이미지 파일

